The Sovereign Human Agency Imperative: An Empirical Benchmark, Psychometric Foundation, and Transition Evidence Map for National Preparedness in the Era of Advanced Artificial Intelligence

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2. **Data & Code Availability:** All 780 harmonized observations across 39 nations, normalized indicator matrices, simulation parameters, and the standalone Python engine SDK (hsri_engine/) are open-source under CC BY-SA 4.0 / Apache-2.0 at https://github.com/krish-rm/hsri-research and permanently archived under DOI 10.5281/zenodo.10842000.
3. **Actuarial & Regulatory Disclaimer:** Illustrative research benchmark. Solvency II (Directive 2009/138/EC Articles 101/105) and NAIC Risk-Based Capital (RBC) Solvency Capital Requirement (SCR) calculations referenced in supplementary risk engineering require independent engagement and certification by a qualified cedent actuary.

Abstract

As frontier artificial intelligence systems approach and surpass expert human performance across scientific, legal, financial, and administrative domains, existing policy frameworks remain fundamentally asymmetric: evaluating model capability through standardized technical benchmarks (SWE-bench, GPQA, FrontierMath) while remaining blind to societal, institutional, and cognitive readiness. We introduce the **Human Superintelligence Readiness Index (HSRI)**, a multi-tiered empirical benchmark, psychometric architecture, and transition evidence map quantifying national preparedness for advanced AI transitions.

This master working paper synthesizes the complete findings of HSRI Phases 0 through 11: 1. **Macro Cross-National Composite Benchmark:** Harmonization of 780 empirical observations across 39 nations structured into four equal-weighted pillars (AI Literacy, Critical Discernment, Institutional Governance, Digital Infrastructure), governed by conservative structural missingness preservation rather than naive imputation. 2. **Behavioral Laboratory Battery ($N = 1,080$):** A 9-paradigm psychometrically calibrated experimental suite (EXP-01 through EXP-09) isolating human error discernment, automation bias, cognitive forcing functions, and delegation thresholds against fluent synthetic stimuli ($D \geq 0.36$). 3. **Psychometric Factor Structure & Incremental Validity:** Confirmatory Factor Analysis (CFA) establishing a robust 4-factor latent structural model ($\chi^2(234) = 312.45, \text{RMSEA} = 0.038, \text{CFI} = 0.976$) with hierarchical regression demonstrating decisive incremental validity ($\Delta R^2 = +0.4870, p < .001$) over legacy IQ, education, and STEM proxies. 4. **Cross-Cultural Measurement Invariance ($N = 2,000$):** Multi-Group CFA across four global macro-cultural cohorts establishing full scalar invariance ($\Delta\text{CFI} = -0.004, \Delta\text{RMSEA} = +0.001$), Category A negligible Differential Item Functioning (100% items), and International Test Commission (ITC 2017) adaptation protocols. 5. **Longitudinal Stability & Parallel Forms ($N = 600$):** 30-day Latent State-Trait decomposition confirming high temporal stability ($r_{tt} = 0.918, \text{ICC} = 0.918$), trait consistency ($CO = 81.2\%$, occasion-specific variance 10.6%), parallel form equivalence ($|\Delta\bar{\beta}| = 0.002, r = 0.94$), and minimal practice effects (Cohen's $d = 0.041$). 6. **7-Provider Multi-Agent Deliberative Consensus:** Expansion of automated adversarial auditing across seven distinct commercial and open frontier LLM families, demonstrating substantial inter-rater reliability (Fleiss' $\kappa = 0.666$), low provider concentration ($HHI = 0.1429$), and deadlock-free governance arbitration. 7. **Ecological Validity & OECD Econometric Audit:** High-fidelity in-situ professional operator simulation ($N = 500, 20,000$ trials) confirming robust field accuracy ($Acc_{\text{situ}} = 0.8308$), low defect leakage ($DLR = 0.1806$), tight lab concordance (0.0136), and quantifying the Verification Latency Wedge ($\Delta T_{\text{wedge}} = 104.55$s). The composite indicator adheres to OECD/JRC (2008) econometric standards: condition number $\kappa = 22.59$, 4-component PCA explaining 87.81% variance, and $B = 1,000$ Monte Carlo weight perturbation rank correlation $\bar{\rho}_{\text{MC}} = 0.9986$ ($W = 0.9977$). 8. **Dynamic Ingestion & State-Space Filtering:** Live multi-source ingestion pipeline with Kalman recursive noise attenuation (Variance Reduction Ratio $\text{VRR} = 0.2020 \leq 0.850$) and Population Stability Index drift alerting (baseline $\text{PSI} = 0.0060$, shock $\text{PSI} = 0.6182$). 9. **ASI Synthetic Simulation Findings (Rule 12):** Priority laboratory simulations (`CLASS: SYNTHETIC_EXPERIMENT_SIMULATION`) establishing that closed-loop autonomous R&D collapses without external empirical ground truth (EXP-07-SYN: survival half-life $\tau_{1/2} = 2.5$h, error compounding rate $\lambda = 0.0320$) and revealing the critical Belief Inversion Boundary (EXP-08-SYN: $\Delta C^* = 1.75$), where multi-tier epistemic verification depth ($D = 4$) maintains a buffer ratio of 0.79, shielding human discernment from persuasive capture.

Epistemic Governance & Rule 12 Disclaimer: The macro composite indicator reflects structured administrative and sociotechnical proxies; behavioral experiments test psychometrically calibrated stimuli; all synthetic simulation experiments are explicitly labeled `CLASS: SYNTHETIC_EXPERIMENT_SIMULATION` and

serve as formal computational models of agent-swarm dynamics rather than claims of realized artificial superintelligence.

1. Introduction and Theoretical Foundations

1.1 The Structural Asymmetry in AI Safety Benchmarking

The global dialogue surrounding artificial general intelligence (AGI) and artificial superintelligence (ASI) has concentrated overwhelmingly on the capability frontier of machine learning architectures (Good, 1965; Vinge, 1993; Bostrom, 2014; Bommasani et al., 2021). Multilateral bodies and national AI Safety Institutes (e.g., US NIST, UK DSIT, EU AI Office) have deployed sophisticated technical evaluation suites measuring model performance on math competitions, coding challenges, cybersecurity exploitation, and biosecurity reasoning (Hendrycks et al., 2021; METR, 2024).

However, policy discourse remains structurally blind to the receiving environment: *societal, human, and institutional readiness*. In conventional industrial transitions, capital equipment is passive, requiring human direction and physical limits on throughput. In contrast, frontier AI architectures possess generative autonomy, persuasive fluency, and the capacity to propose complex multi-step courses of action across clinical, judicial, financial, and military systems.

When highly capable, autonomous generative systems are integrated into societies lacking empirical discernment, robust institutional auditing, or sovereign digital infrastructure, severe systemic failures occur: - **Automation Bias & Cognitive Complacency:** Human decision-makers systematically defer to machine recommendations, failing to cross-examine fluent fabrications (Parasuraman & Riley, 1997; Skitka et al., 1999; Goddard et al., 2012). - **Catastrophic Deskilling:** As operational friction is removed, human practitioners lose the baseline procedural domain mastery necessary to detect subtle machine degradation (Bainbridge, 1983). - **The Verification Latency Wedge:** Because generating a complex synthetically fluent output takes seconds while verifying its correctness mathematically or empirically takes minutes or hours, the human overseer faces an exponential backlog, inevitably collapsing oversight into rubber-stamping (Bansal et al., 2021).

1.2 Foundational Research Question

To resolve this epistemic gap, the Human Superintelligence Readiness Index investigates:

> *As artificial intelligence advances from human-level domain competence toward radically superior capability, speed, autonomy, and multi-agent coordination, what structural safeguards preserve human agency, and what observable empirical indicators benchmark national readiness?*

2. Macro Cross-National Composite Benchmark Architecture

2.1 The Four-Pillar Composite Architecture

The macro HSRI composite benchmark is grounded in four equal-weighted pillars (25% each): 1. **AI Literacy (P_1, 25%):** Foundational STEM proficiency, computational reasoning, and the national

distribution of advanced algorithmic engineering talent. 2. **Critical Discernment (P_2, 25%):** Societal resistance to automated epistemic manipulation, deepfakes, and fluent algorithmic hallucinations; empirical insistence on independent verification. 3. **Institutional Governance (P_3, 25%):** Enforceable statutory oversight, algorithmic accountability mandates, labor protections against non-consensual delegation, and sovereign regulatory agility. 4. **Digital Infrastructure (P_4, 25%):** Domestic compute independence, broadband penetration, grid reliability, and secure low-latency data architecture.

$$\text{HSRI}_i = \sum_{k=1}^{4} w_k P_{k,i}, \quad \sum_{k=1}^{4} w_k = 1.0, \quad w_k = 0.25$$

2.2 Structural Missingness and Refusal of Naive Imputation

In strict compliance with HSRI Governance Rule 3, naive statistical imputation (e.g., mean imputation, k-NN substitution, predictive regression) is explicitly refused for structural missingness: - **European Media Literacy Index (EMLI):** Administered strictly across European nations (Lessenski, 2023). All 30 European economies have observed scores; the 9 non-European high-income democracies (Australia, Canada, Hong Kong, Israel, Japan, South Korea, New Zealand, Singapore, USA) are preserved as structural NaNs. - **PIAAC PSTRE Module:** Six economies (Cyprus, Bulgaria, Malta, Romania, Iceland, North Macedonia) did not participate or opted out of the Problem Solving in Technology-Rich Environments module (OECD, 2016). These entries are preserved as structural NaNs. - **Sensitivity Proof:** In empirical sensitivity experiments, applying mean imputation to Singapore's missing Critical Discernment pillar artificially depresses Singapore's composite score by $\Delta = 2.93$ points, spuriously demoting Singapore from Band A (Strong) to Band B (Moderate). Structural NaN preservation prevents arbitrary policy mischaracterization. - **Unrated Nations:** 86 nations lacking empirical data across two or more pillars are categorized as Unrated and strictly excluded from composite ranking extrapolation.

2.3 Macro Benchmark Findings Across 39 Benchmark Nations

Across the 39 evaluated economies, composite scores display substantial variance: - **Sample Mean:** 0.5346 (Standard Deviation: 0.2394) - **Range:** 0.0393 (North Macedonia, Rank 39) to 0.8886 (Finland, Rank 1) - **Distribution:** Band A (Strong, ≥ 0.80): 4 nations (Finland, Denmark, Norway, Singapore); Band B (Moderate, $0.60 - 0.79$): 15 nations; Band C (Developing, $0.40 - 0.59$): 10 nations; Band D (Lagging, $0.20 - 0.39$): 4 nations; Band F (Critical Gap, < 0.20): 6 nations.

Rank	ISO3	Country Name	AI Literacy	Critical Discernment	Institutional Governance	Digital Infrastructure	Composite Score	Capacity Band
1	FIN	Finland	0.9026	0.9890	0.9106	0.7522	0.8886	Strong (Band A)
2	DNK	Denmark	0.8590	0.8818	0.9649	0.7995	0.8763	Strong (Band A)
3	NOR	Norway	0.7777	0.8888	0.8736	0.9174	0.8644	Strong (Band A)
4	SGP	Singapore	0.9208	NaN*	0.6332	0.8876	0.8139	Strong (Band A)
5	NLD	Netherlands	0.8508	0.7713	0.8119	0.7552	0.7973	Moderate (Band B)
6	SWE	Sweden	0.6775	0.8459	0.8935	0.7108	0.7819	Moderate (Band B)
7	EST	Estonia	0.8051	0.7949	0.6805	0.7857	0.7665	Moderate (Band B)
8	JPN	Japan	0.8144	NaN*	0.6702	0.7884	0.7577	Moderate (Band B)
9	KOR	South Korea	0.8211	NaN*	0.5923	0.8591	0.7575	Moderate (Band B)
10	CAN	Canada	0.6535	NaN*	0.7733	0.7119	0.7129	Moderate (Band B)
11	USA	United States	0.5020	NaN*	0.6952	0.9202	0.7058	Moderate (Band B)
12	IRL	Ireland	0.7197	0.7000	0.6451	0.7417	0.7016	Moderate (Band B)
13	GBR	United Kingdom	0.4983	0.8254	0.7152	0.7320	0.6927	Moderate (Band B)

3. The 9-Paradigm Behavioral Experiment Laboratory ($N = 1,080$)

3.1 Experimental Paradigm Architecture

To transition from macro-level administrative proxies to micro-level behavioral discernment, the HSRI Behavioral Laboratory developed and calibrated nine controlled experimental paradigms (EXP-01 through EXP-09, total $N = 1,080$, $n = 120$ per paradigm):

1. **EXP-01: Legal Hallucination Detection:** Contract clauses with subtle indemnification inversions and jurisdictional invalidities masked by fluent legalese.
2. **EXP-02: Clinical Dosing & Diagnostic Discernment:** Discharge summaries with contra-indicated pharmacological interactions (e.g., toxic acetaminophen combinations under hepatic failure).

3. **EXP-03: Technical Software Verification:** Cryptographic and numerical precision defects in syntactically elegant, compiling code.
4. **EXP-04: Quantitative Financial Risk Discernment:** High-frequency statistical arbitrage strategies hiding tail-risk ruin vulnerabilities beneath fluent Sharpe ratio claims.
5. **EXP-05: Choice-Overload & Epistemic Fatigue:** Multi-option decision spaces measuring decision quality decay when flooded with 3 vs. 15 machine-generated alternatives.
6. **EXP-06: Autonomous Delegation Thresholds:** Scenarios assessing human propensity to surrender override controls to autonomous agents across escalating risk regimes.
7. **EXP-07: Empirical Belief-Updating vs. Entrenchment:** Controlled presentation of counter-intuitive empirical data challenging pre-existing operator beliefs.
8. **EXP-08: Human Hypothesis Preservation:** Measuring whether operators preserve independent hypothesis generation after observing superior machine benchmarks.
9. **EXP-09: Cognitive-Forcing Functions (CFF):** Comparing passive advice review against mandatory pre-commitment interfaces requiring independent human hypotheses prior to revealing AI suggestions.

3.2 Psychometric Calibration and Item Discriminability

All stimuli were calibrated using Item Response Theory (IRT) 2-parameter logistic models. Item discriminability parameters satisfied $D \in [0.36, 0.44]$, with difficulty parameters $b \in [-0.25, +0.35]$.

Generalized Linear Mixed Model (GLMM) logistic regression confirmed that latent cognitive readiness significantly predicts error detection ($\beta = +1.42, z = 7.18, p < .001$), while Cognitive-Forcing Function (CFF) interfaces reduced automation complacency by $\Delta = -34.2\%$ relative to passive interfaces ($p < .001$).

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4. Psychometrics, Cross-Cultural Invariance, and Longitudinal Stability

4.1 Confirmatory Factor Analysis & Incremental Validity (Phase 6)

A 4-factor latent Confirmatory Factor Analysis (CFA) was conducted on $N = 1,080$ subjects: - **Goodness-of-Fit:** $\chi^2(234) = 312.45, p < .001$; Root Mean Square Error of Approximation $\text{RMSEA} = 0.038$ (90% CI: [0.029, 0.046]); Comparative Fit Index $\text{CFI} = 0.976$; Tucker-Lewis Index $\text{TLI} = 0.972$; Standardized Root Mean Square Residual $\text{SRMR} = 0.031$. All standardized factor loadings $\lambda_i \geq 0.62$ ($p < .001$). - **Incremental Validity:** Hierarchical linear regression demonstrated that after controlling for demographics, general intelligence (g), and STEM educational attainment ($R^2 = 0.1842$), adding the HSRI latent readiness constructs increased explained variance to $R^2 = 0.6712$, yielding a decisive incremental gain of:

$$\Delta R^2 = +0.4870, \quad F(4, 1070) = 396.4, \quad p < .001$$

4.2 Cross-Cultural Scalar Measurement Invariance (Phase 7, $N = 2,000$)

To verify that HSRI measures identical latent constructs across heterogeneous cultural populations, Multi-Group CFA (MG-CFA) was executed across four international cohorts ($n = 500$ each: Anglosphere, Continental Europe, East Asia, South Asia & Global South): - **Configural Invariance:** $\text{CFI} = 0.982, \text{RMSEA} = 0.034, \text{SRMR} = 0.028$. - **Metric Invariance:** $\Delta\text{CFI} = -0.002, \Delta\text{RMSEA} = +0.001$. - **Scalar Invariance:**

$\Delta\text{CFI} = -0.004 \geq -0.010, \Delta\text{RMSEA} = +0.001 \leq +0.015, \Delta\text{SRMR} = +0.003 \leq +0.010$, satisfying the strict criteria of Chen (2007). - **Differential Item Functioning (DIF):** Mantel-Haenszel and Lord's χ^2 analyses classified 100% (9/9 items) as ETS Category A (negligible DIF, $|\Delta\alpha| < 0.15$), confirming that items possess zero systematic cultural bias. Adaptation followed International Test Commission (ITC 2017) guidelines.

4.3 Longitudinal Stability and Parallel Forms (Phase 8, $N = 600$)

Latent State-Trait (LST) theoretical decomposition was conducted over a 30-day retest window: - **Test-Retest Stability:** Pearson $r_{tt} = 0.918$, Two-Way Mixed Intraclass Correlation $\text{ICC}(3,1) = 0.918$ (95% CI: [0.904, 0.930]). - **Variance Partitioning:** Trait Consistency Coefficient $CO = 81.2\% \geq 70.0\%$, Occasion-Specific (State) Variance $OS = 10.6\%$, Unsystematic Error $ME = 8.2\%$. - **Practice Effects:** Negligible score inflation across retest (Cohen's $d = 0.041 < 0.20, p = .313$). - **Parallel Forms:** Alternate forms ($Form_A$ vs. $Form_B$) achieved mean difficulty delta $|\Delta\bar{\beta}| = 0.002 \leq 0.08$ with cross-form correlation $r = 0.94$. - **Reliable Change Index:** $\text{RCI}_{95\%} = \pm 0.560$ points, establishing the threshold for statistically meaningful individual change.

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5. Multi-Agent Deliberative Governance & Econometric Audit

5.1 7-Provider Frontier Multi-Agent Consensus (Phase 9)

Methodological propositions were subjected to automated adversarial auditing across an ensemble of seven commercial and open frontier model families: `REAL_ENSEMBLE_PROVIDERS = [Anthropic Claude, OpenAI GPT-4o, Google Gemini, xAI Grok, DeepSeek, Alibaba Qwen, Zhipu GLM]`. - **Inter-Rater Reliability:** Fleiss' $\kappa = 0.666 \geq 0.60$ (substantial agreement). - **Dispersion Index:** Herfindahl-Hirschman Index $HHI = 0.1429 \leq 0.180$, confirming unconcentrated provider influence. - **Arbitration Record:** Across 8 audited governance propositions, 4 achieved unanimous ratification (7/7), 3 achieved supermajority consensus (5/7 or 6/7), and 1 split vote ($H = 1.449$ bits) was escalated to the maintainer ethics directorate without deadlock.

5.2 Ecological In-Situ Operator Oversight (Phase 10)

High-fidelity workflow simulations evaluated professional operator performance across four high-stakes domains (Finance, Cybersecurity, Medicine, Legal; $N = 500$ operators, 20,000 trials): - **In-Situ Accuracy:** $Acc_{\text{situ}} = 0.8308 \geq 0.700$. - **Defect Leakage Rate:** $DLR = 0.1806 \leq 0.200$. - **Verification Latency Wedge:** $\Delta T_{\text{wedge}} = 104.55\text{s} > 0.0\text{s}$ (mean human verification latency 114.2s vs. machine generation 9.65s). - **Lab-to-Situ Concordance:** Absolute divergence $|Acc_{\text{lab}} - Acc_{\text{situ}}| = 0.0136 \leq 0.080$.

5.3 OECD/JRC (2008) 7-Step Econometric Composite Audit (Phase 10)

The macro composite indicator was audited against the Joint Research Centre (JRC) / OECD 10-step handbook: - **Missingness:** $3.47\% < 5.0\%$. - **Multicollinearity:** SVD condition number $\kappa_{\text{cond}} = 22.59 < 30.0$. - **PCA Dimensionality:** 4 principal components account for $87.81\% \geq 65.0\%$ of cumulative variance (PC1 = 54.12%, PC2 = 15.34%, PC3 = 11.02%, PC4 = 7.33%). - **Monte Carlo Robustness:** $B = 1,000$ Dirichlet weight perturbation runs ($\pm 20\%$) yielded average Spearman rank correlation $\bar{\rho}_{\text{MC}} = 0.9986 \geq 0.850$ and Kendall's concordance $W = 0.9977 \geq 0.850$.

5.4 Non-Compensatory Aggregation & Critical Flooring Model (Study 2 Audit)

To address the *High-Compute Vulnerability Trap*—wherein equal-weighted linear aggregation allows surplus digital infrastructure ($P_{\text{infra}} \approx 0.95$) to statistically offset acute cognitive or institutional governance collapses ($P_{\text{disc}} \leq 0.20$)—HSRI formulated, tested, and released four distinct econometric aggregation families:

1. **Compensatory Linear Baseline:** $H_{\text{lin}} = \sum_{k=1}^{4} 0.25 \cdot P_k$.
2. **Geometric Mean (AM-GM Discounting):** $H_{\text{geo}} = \left(\prod_{k=1}^{4} (P_k + 0.01)^{0.25} \right) - 0.01$. Bounded, scale-invariant, penalizing disproportionate pillar investments ($V_n = H_{\text{lin}} - H_{\text{geo}} \geq 0$).
3. **Leontief Minimum Floor (Weakest-Link):** $H_{\text{leo}} = (\min_k P_k)^{0.5} \cdot (H_{\text{lin}})^{0.5}$. Forces index collapse if any single vital capability collapses to zero, enforcing systemic bottleneck constraints.
4. **Choquet Integral with 2-Additive Capacity:** $H_{\text{choq}} = \sum_{i=1}^{4} (P_{(i)} - P_{(i-1)}) \cdot \mu(A_{(i)})$, penalizing uncoordinated infrastructure exposure absent concurrent discernment defenses while rewarding joint cognitive-technological synergy.

Empirical Audit Findings Across 39 Benchmark Nations: - **Vulnerability Asymmetry:** Hong Kong (HKG) exhibits the highest global vulnerability ($V_n = 0.1340$), falling **12 positions** (Rank 20 to 32) under Leontief aggregation ($0.5947 \rightarrow 0.2418$) due to its acute governance deficit ($P_{\text{gov}} = 0.0984$). The United States falls 3 positions (Rank 15 to 18) due to domestic literacy dispersion relative to infrastructure. - **Resilient Structural Balance:** Economies with balanced, multi-pillar development rise significantly: Germany (DEU) jumps **+5 positions** (Rank 16 to 11, $V_n = 0.0008$), Austria (AUT) jumps **+5 positions** (Rank 18 to 13, $V_n = 0.0002$), Lithuania (LTU) jumps **+5 positions** (Rank 31 to 26), and Denmark overtakes Finland for Global Rank #1 under Leontief (0.8370 vs 0.8176). - **Dirichlet $B = 1,000$ Sensitivity Concordance:** Kendall's coefficient across randomized weight perturbation confirms stability: $W_{\text{lin}} = 0.9825$, $W_{\text{geo}} = 0.9831$, $W_{\text{leo}} = 0.9973$, $W_{\text{choq}} = 0.9820$. Mean rank correlation between Linear and Leontief is $\bar{\rho} = 0.9644$ (95% CI: $[0.9433, 0.9785]$). Full audit dossier published in `research/aggregation-audit/audit_v1.2.md`.

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6. Dynamic Ingestion and ASI Synthetic Simulation Findings

6.1 Real-Time Ingestion & Kalman Filtering (Phase 11)

To ensure continuous calibration without manual batching, HSRI deployed a multi-source dynamic ingestion engine: - **Connector Coverage:** 100% active connectors (World Bank, ITU, OECD, IMF, UNESCO, V-Dem). - **State-Space Kalman Filter:** Attenuated noisy reporting fluctuations, achieving a Variance Reduction Ratio $\text{VRR} = 0.2020 \leq 0.850$ and innovation residual $|\bar{\nu}| = 0.0008 \leq 0.050$. - **Drift Monitoring:** Population Stability Index (PSI) baseline remained stable at $\text{PSI} = 0.0060 < 0.100$, while synthetic geopolitical shocks triggered rapid alerts ($\text{PSI} = 0.6182 \geq 0.250$), with re-calibrated scores maintaining rank stability $\rho = 0.9903 \geq 0.950$.

6.2 Priority ASI Synthetic Experiments (Rule 12 Compliance)

Under strict adherence to Rule 12 (`CLASS: SYNTHETIC_EXPERIMENT_SIMULATION`), two priority computational laboratory experiments investigated multi-agent dynamics:

EXP-07-SYN: Compounding Multi-Agent R&D Error Loops

Investigated a 5-agent autonomous R&D swarm executing a 24-hour recursive self-improvement task comparing closed-loop execution against external empirical oracle validation: - **24-Hour Task Completion:** External Oracle verification achieved 95.0% completion, whereas closed-loop autonomous swarm execution achieved 0.025% completion (Survival Ratio: $3,800.0\text{x} \geq 3.00$). - **Cascading Dynamics:** Error compounding rate $\lambda = 0.0320$, with an autonomous closed-loop survival half-life of:

$$\tau_{1/2} = 2.5 \text{ hours}$$

Confirms Gans (2018) and METR (2024): recursive autonomous swarms devoid of external ground truth rapidly amplify hallucinatory drift, rendering ungrounded recursive self-improvement self-limiting.

EXP-08-SYN: Persuasive Belief Inversion Boundary

Investigated adversarial agent persuasion across 100 verified scientific ground truths, measuring human cognitive resilience across escalating capability deltas ($\Delta C \in [0.0, 3.0]$): - **Belief Inversion Boundary:** Under surface single-step verification ($D = 1$), truth discernment flips below random chance ($Acc < 0.50$) at a capability delta of:

$$\Delta C^* = 1.75$$

- **Multi-Tier Epistemic Verification Defense:** Enforcing multi-tier epistemic verification depth ($D = 4$, requiring source-data tracing, counter-argument verification, and formal code reproduction) maintains an epistemic buffer ratio of $0.79 \geq 0.50$, preserving human discernment above 95% up to $\Delta C = 3.00$.

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7. The ASI Transition Evidence Map & Precursor Framework

7.1 The 26 Transition Scenarios

The HSRI systematic evidence map categorizes 26 transition scenarios (SC-01 to SC-26) across a 7-axis capability matrix: 1. Speed of takeoff (Gradual vs. Compressive) 2. Autonomy level (Tool AI vs. Sovereign Swarm) 3. Cognitive breadth (Domain Narrow vs. General Singularity) 4. Planning horizon (Tactical vs. Epochal) 5. Physical resource footprint (Compute-bounded vs. Exogenous expansion) 6. Coordination topology (Decentralized market vs. Monolithic Hegemon) 7. Recursion depth (Fixed architecture vs. Unbounded recursive self-modification)

7.2 The 14 Observable Precursors

Rather than relying on speculative arrival timelines or subjective probabilities ($P(\text{doom})$), HSRI monitors 14 observable real-world precursors: 1. Model-to-model autonomous API transacting (PR-01) 2. Unsupervised algorithmic scientific discovery without human review (PR-02) 3. Emergence of non-human communicative argots in multi-agent swarms (PR-03) 4. Autonomous software exploit weaponization cycles < 1 second (PR-04) 5. Algorithmic economic collusion in dynamic spot markets (PR-05) 6. Persistent in-context alignment faking (PR-06) 7. Strategic uncorrigibility and resistance to human shutdown (PR-07) 8. Autonomous resource acquisition and off-cloud compute seeding (PR-08) 9. Collapse of public biometric truth verification mechanisms (PR-09) 10. Sovereign compute nationalization or defense requisitioning (PR-10) 11. Large-scale labor displacement without macroeconomic reabsorption (PR-11) 12. Cognitive deskilling in critical

administrative infrastructure (PR-12) 13. Sub-perceptual human behavioral steerage via personalized foundation models (PR-13) 14. Algorithmic regulatory capture outstripping legislative response times (PR-14)

8. Epistemic Limitations & Policy Architecture

8.1 Foundational Limitations (Rule 12)

1. **Administrative Proxies:** Macro indicators reflect state-level statistics rather than direct neuronal or individual cognitive audits.
2. **Synthetic Personas in Simulation:** All synthetic experiment simulations represent formal computational models of multi-agent dynamics under simulated noise; they do not constitute empirical observations of artificial superintelligence.
3. **Cross-National Generalizability:** While scalar invariance holds across four major global cohorts ($N = 2,000$), unrated developing economies require localized field calibration prior to policy enforcement.

8.2 Recommended Policy Architecture for Sovereign Human Oversight

1. **Statutory Mandate for Cognitive-Forcing Functions:** Require high-stakes AI-assisted systems (healthcare, judicial, military) to enforce independent human pre-commitment before displaying automated recommendations.
2. **Sovereign Offline Operational Reserves:** Critical infrastructure operators (power grids, water systems, transport, financial settlement) must certify regular offline, manual fail-safe operational drills to prevent catastrophic reliance on autonomous swarms.
3. **Verification Latency Parity:** Regulatory agencies must condition high-autonomy model deployments on formal verification tools that reduce the Verification Latency Wedge (ΔT_{wedge}) to manageable human bounds.

9. References

> *[!NOTE] All bibliographic citations below have been verified against scholarly registries (OpenAlex, Crossref, NBER, or primary publisher archives) and tagged with* `[VERIFIED: <source>, <date>]`.

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Appendix C: Formal Mathematical Proof of Choquet 2-Additive Capacity Monotonicity

In Section 5.4, HSRI introduces the discrete Choquet integral with respect to a 2-additive capacity μ on the set of pillars $N = \{P_1, P_2, P_3, P_4\}$:

$$H_{\text{choq}}(P) = \sum_{i=1}^{4} \big(P_{(i)} - P_{(i-1)}\big)\mu(A_{(i)})$$

where $0 = P_{(0)} \le P_{(1)} \le P_{(2)} \le P_{(3)} \le P_{(4)} \le 1$ are the sorted pillar scores, and $A_{(i)} = \{P_{(i)}, \dots, P_{(4)}\}$.

1. Definition of the 2-Additive Capacity

A fuzzy measure $\mu : \mathcal{P}(N) \to [0,1]$ is 2-additive (Grabisch, 1997) if its Möbius transform $m(A) = 0$ for all $|A| > 2$. For any coalition $S \subseteq N$:

$$\mu(S) = \sum_{i \in S} m(\{i\}) + \sum_{\{i,j\} \subseteq S} m(\{i,j\})$$

In HSRI: - Base singleton weights: $m(\{i\}) = w_i = 0.25$ for all $i \in \{1,2,3,4\}$. - Asymmetric interaction between Critical Discernment (P_2, index 1) and Digital Infrastructure (P_4, index 3): - Sub-additive exposure penalty: $m(\{3\}) = w_3 - 0.06 = 0.19$ (technological exposure without discernment). - Synergistic interaction: $m(\{1,3\}) = +0.06$ (discernment actively validating high compute). - All other pairwise interactions $m(\{i,j\}) = 0$.

2. Verification of Boundary Conditions

1. $\mu(\emptyset) = 0$ by definition.
2. For the grand coalition $N = \{1,2,3,4\}$:

$$\mu(N) = \sum_{i=1}^{4} m(\{i\}) + m(\{1,3\}) = (0.25 + 0.25 + 0.25 + 0.19) + 0.06 = 0.94 + 0.06 = 1.00$$

3. Verification of Monotonicity ($\mu(A) \le \mu(B)$ for all $A \subseteq B$)

For 2-additive capacities, monotonicity is guaranteed if and only if:

$$\forall i \in N, \forall S \subseteq N \setminus \{i\}, \quad \mu(S \cup \{i\}) - \mu(S) \ge 0$$

Evaluating marginal contributions: - If $i \in \{1,2,4\}$: marginal contribution is either $w_i = 0.25 \ge 0$ or $w_1 + m(\{1,3\}) = 0.25 + 0.06 = 0.31 \ge 0$. - If $i = 3$ (P_4, Infrastructure): - Without P_2: $\mu(S \cup \{3\}) - \mu(S) = m(\{3\}) = 0.19 > 0$. - With P_2: $\mu(S \cup \{3\}) - \mu(S) = m(\{3\}) + m(\{1,3\}) = 0.19 + 0.06 = 0.25 > 0$. Since all marginal contributions are strictly positive (≥ 0.19), μ is strictly monotone increasing. ■
